

GIS-Based Assessment of Rooftop Rainwater Harvesting Potential in the Colombo District, Sri Lanka

A Building-Scale Analysis of 234,927 Structures Using Open Geospatial and Climate Data

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Abstract

Rapid urbanisation and seasonal water stress make rooftop rainwater harvesting (RWH) an attractive supplementary water source for Sri Lanka's Western Province. This study quantifies the building-scale RWH potential of the entire Colombo District by combining open building-footprint data from OpenStreetMap, official administrative boundaries from GADM, and eleven years (2015–2025) of daily rainfall from the Open-Meteo Historical Weather API. Roof areas were computed for **234,927 buildings** across twelve Divisional Secretariat (DS) divisions using a metric national grid, and annual harvest was estimated from roof area, mean annual rainfall (2,835 mm), and roof-material runoff coefficients. The district shows a combined potential of approximately **78,292 megalitres per year (78.3 GL)**, with 74.4% of buildings classified as High potential ($\geq 150,000$ L/yr). Kaduwela, Dehiwala-Mount Lavinia, and Moratuwa rank highest. The results provide a practical, replicable evidence base for municipal water planning and household-level RWH promotion.

Keywords: *rainwater harvesting; GIS; OpenStreetMap; runoff coefficient; urban water management; Colombo; Sri Lanka.*

1. Introduction

Sri Lanka receives abundant rainfall, yet the Western Province — home to the Colombo District and the country's densest urban population — regularly experiences localised water-supply pressure, stormwater flooding, and rising demand on the centralised pipe network. Rooftop rainwater harvesting (RWH) offers a decentralised, low-cost intervention that simultaneously supplements domestic supply and reduces peak stormwater runoff. However, planning RWH at a city or district scale requires knowing how much water the existing building stock could realistically capture, and where that potential is concentrated.

Traditional RWH assessments rely on manual surveys or coarse land-use assumptions, which are costly and quickly outdated. This study takes an open-data, building-scale approach: every mapped building in the Colombo District is treated as an individual catchment, and its annual harvest is estimated from its measured roof area, the local long-term rainfall record, and a material-specific runoff coefficient. The objectives are (i) to quantify the total and per-building RWH potential across all twelve DS divisions of the district, (ii) to classify each building into High, Medium, or Low potential tiers, and (iii) to produce ranked, map-based outputs that municipal planners and households can act upon.

2. Literature Review

[This section is reserved for the literature review. Suggested structure and prompts are provided below to be expanded before submission.]

- *Global and regional RWH practice — potential estimation methods and adoption rates in tropical monsoon climates.*
- *Runoff-coefficient literature — FAO (2010), WHO/UNEP, and Sri Lanka NBRO guidance on roof-material collection efficiency.*
- *GIS and open geospatial data (OpenStreetMap) for building-scale hydrological modelling.*
- *Sri Lankan policy context — the National Rainwater Harvesting Policy and Western Province urban water studies.*

(Approx. 3/4 page reserved.)

3. Methodology

The analysis followed a reproducible two-notebook Python pipeline: (1) data collection, and (2) preprocessing, calculation, and visualisation. All processing used open, freely licensed data and no proprietary GIS software.

3.1 Data Sources

Dataset	Source / Provider	Role in the study
Building footprints	OpenStreetMap (Overpass API)	Roof polygons and roof-material tags
Administrative boundaries	GADM v4.1 (level-2 DS divisions)	Exact non-overlapping city polygons
Daily rainfall 2015–2025	Open-Meteo Historical Weather API	Mean annual rainfall for the formula

Table 1. Open datasets used, with their role in the RWH pipeline.

Buildings were queried using each DS division's exact GADM polygon (Overpass *poly*: filter) rather than a bounding box, so that every building belongs to exactly one division and no cross-city duplication occurs. Rainfall was collected for Colombo (6.9271° N, 79.8612° E), giving a mean annual total of **2,835 mm** over the 2015–2025 period.

3.2 Preprocessing

- Invalid geometries repaired; roof area computed in EPSG:5235 (Sri Lanka metric grid).
- Buildings outside 10–10,000 m² removed as unrealistic; raw records reduced from 236,894 to 234,927 (1,967 removed).
- Roof materials normalised to four canonical classes; roofs with no defined runoff coefficient (e.g. cadjan/straw, glass) excluded.

3.3 RWH Formula and Classification

Annual harvest for each building was estimated as:

Annual Harvest (L) = Roof Area (m ²) × Annual Rainfall (m) × Runoff Coefficient × 1,000			
Roof Type	Runoff Coeff. (C)	Buildings (n)	Share
Asbestos	0.70	170,383	72.5%
Concrete	0.85	48,056	20.5%
Tile	0.80	13,943	5.9%
Metal Sheet	0.90	2,545	1.1%

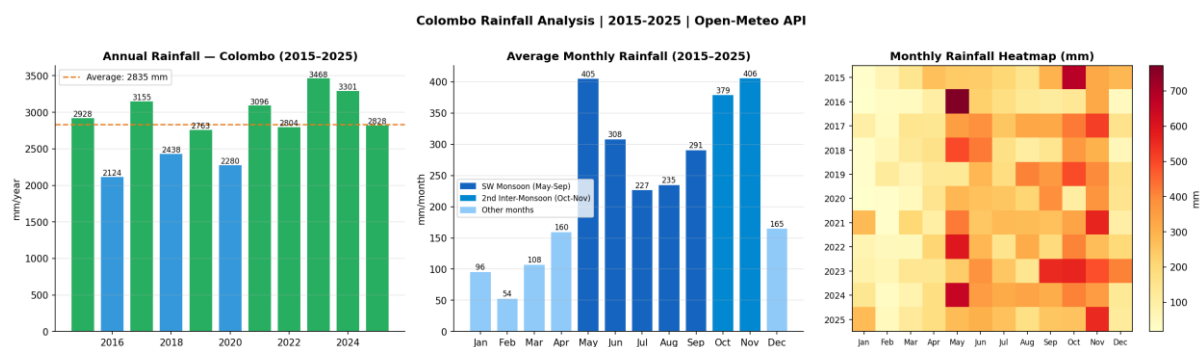
Table 2. Roof-material runoff coefficients (FAO 2010; WHO/UNEP; Sri Lanka NBRO) and building counts.

Each building was then classified by annual yield: **High** ≥ 150,000 L/yr, **Medium** 50,000–149,999 L/yr, **Low** < 50,000 L/yr. A recommended storage-tank size was also derived from the peak-rainfall month, and results were rendered as static charts and an interactive Folium web map.

4. Results

4.1 Rainfall Regime

Colombo's rainfall is strongly bimodal, driven by the South-West monsoon (May–September) and the second inter-monsoon (October–November). May and November are the wettest months (~405 mm each), while February is driest (~54 mm). The eleven-year mean of 2,835 mm/yr anchors the harvest calculation.



4.2 Building Stock and Roof Materials

The building stock is dominated by asbestos-sheet roofs (72.5%), followed by concrete (20.5%). The median roof area is 117 m² and the mean 159 m², with a right-skewed distribution typical of dense residential settlement. Kaduwela, Dehiwala-Mount Lavinia, and Moratuwa hold the largest building counts.

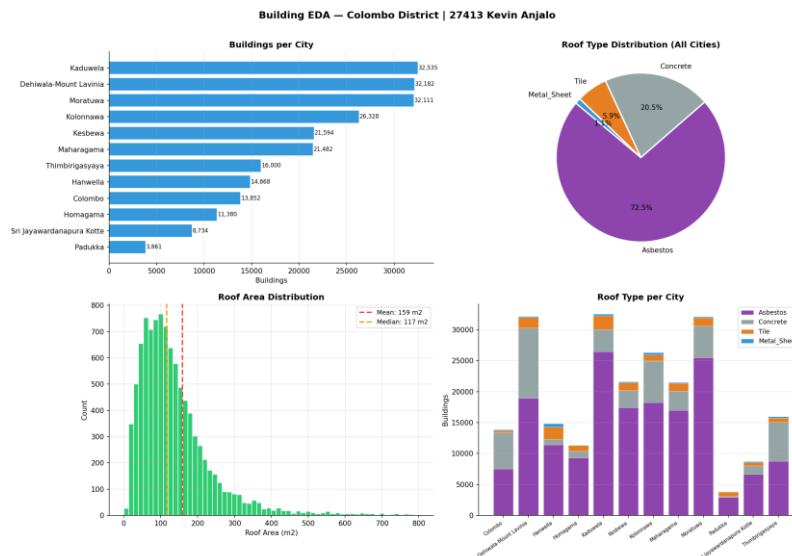


Figure 2. Building EDA — buildings per city, roof-type distribution, roof-area histogram, and roof type by city.

4.3 District RWH Potential

Across all 234,927 buildings, the Colombo District has a combined potential of **78,292 ML/yr (78.3 GL)**. The mean per-building yield is 333,262 L/yr (median 242,244 L/yr), and the median recommended tank size is 35,000 L. **74.4%** of buildings fall in the High tier, 22.7% Medium, and only 2.9% Low — confirming that the great majority of the existing roofscape is well suited to RWH.

DS Division	Buildings	Avg Roof (m²)	Harvest (ML/yr)
Kaduwell	32,535	153	10,248.4
Dehiwala-Mount Lavinia	32,182	148	10,226.2
Moratuwa	32,111	140	9,288.7
Maharagama	21,482	176	7,817.1
Kolonnawa	26,328	137	7,592.7
Colombo	13,852	238	7,189.9
Thimbrigasyaya	16,000	205	7,124.4
Kesbewa	21,594	137	6,085.0
Hanwella	14,868	140	4,323.1
Homagama	11,380	171	3,986.4
Sri Jayawardanapura Kotte	8,734	172	3,107.8
Padukka	3,861	162	1,302.6

Table 3. RWH potential by DS division, ranked by total annual harvest.

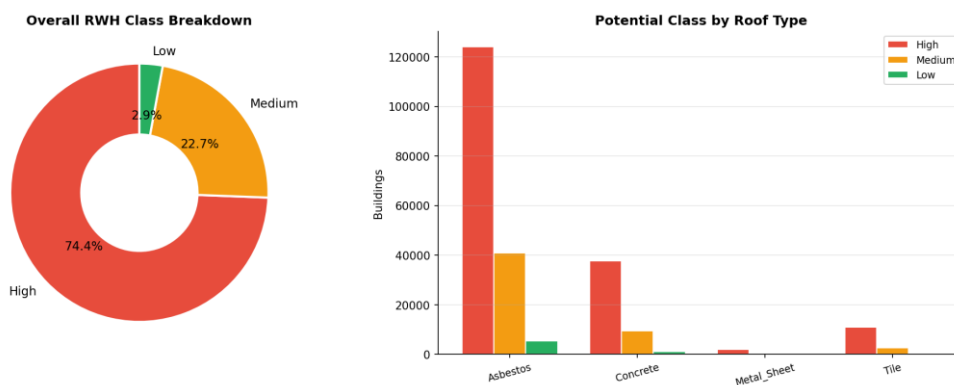


Figure 4. Overall RWH class breakdown and potential class by roof type.

The map-based output (rwh_colombo_map.html) renders each building coloured by potential tier with a harvest-intensity heatmap, allowing planners to locate high-yield clusters visually. High-potential buildings are heavily

concentrated on asbestos and concrete roofs, which together account for the overwhelming majority of the district's harvestable volume.

5. Limitations

- OSM completeness varies — roof-material tags are inferred where missing, and unmapped buildings are not counted, so district totals are conservative.
- A single Colombo rainfall station represents the whole district; localised orographic variation (e.g. Hanwella, Padukka) is not captured.
- The model estimates theoretical supply potential, not demand-matched yield; first-flush losses, tank overflow, and household consumption are not modelled.
- Roof area is taken as the horizontal footprint; pitched-roof surface area and shared/multi-storey ownership are simplifications.

6. Conclusion

Using only open geospatial and climate data, this study demonstrates that the Colombo District's existing building stock could theoretically capture on the order of 78 gigalitres of rainwater per year, with nearly three-quarters of buildings offering High potential. The building-scale, GIS-based method is transparent, low-cost, and directly reproducible for other districts. The ranked city and map outputs give municipal authorities an actionable basis for prioritising RWH incentives — beginning with the high-density, high-yield divisions of Kaduwela, Dehiwala-Mount Lavinia, and Moratuwa. Future work should couple this supply estimate with household water demand and cost–benefit analysis to convert potential into implementable policy.

Data Sources & Software

- OpenStreetMap contributors — building footprints via the Overpass API (openstreetmap.org, ODbL).
- GADM v4.1 — Database of Global Administrative Areas, level-2 boundaries (gadm.org).
- Open-Meteo — Historical Weather API, daily precipitation 2015–2025 (open-meteo.com).
- Runoff coefficients: FAO (2010); WHO/UNEP guidance; Sri Lanka National Building Research Organisation (NBRO).
- Software: Python — GeoPandas, Shapely, Pandas, NumPy, Matplotlib, Folium.